

Dynamic Gravity (MR): Core Master Equation Sheet

Native Spectral-Projection Framework

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Purpose and Status

This sheet consolidates the reduced Dynamic Gravity equation framework into one native MR form. The organizing principle is that the validated observable sectors are projections of a single internal spectral system. This version intentionally focuses on the core gravity, Kerr, finite-core, gauge, Higgs, gauge-Higgs closure, gravitational-wave, and validation-test equations.

The status labels used below are:

D = derived from MR structure, N = normalization/convention, C = constrained by consistency or data,

1 Master Chain

$$\boxed{\Psi \rightarrow \hat{O}_\Psi \rightarrow \{u_n, \mu_n\} \rightarrow P_i[\Psi] \rightarrow \mathcal{O}_i} \quad (1)$$

The universal flow is: master projection field, internal spectral operator, eigenmodes, sector projection, and observable.

$$\boxed{\mathcal{O}_{\text{phys}} = \mathcal{P} \left[\Psi, \hat{O}_\Psi, \{u_n, \mu_n\} \right]} \quad (2)$$

$$\boxed{\Psi \rightarrow \hat{O}_\Psi \rightarrow \{u_n, \mu_n\} \rightarrow \{P_g, P_{\text{rot}}, P_H, P_A, P_X\} \rightarrow \{g_{\mu\nu}, g_{\mu\nu}^{\text{Kerr}}, H, A_\mu, \mathcal{O}_{\text{GW}}\}} \quad (3)$$

2 Internal Hilbert Space, Action, and Spectral System

Equation	Meaning	Tag
$\mathcal{H}_w = L^2(\mathbb{R}, dw)$	(4) Internal projection coordinate as a Hilbert-space coordinate.	D

Equation	Meaning	Tag
$\langle f g\rangle = \int_{\mathbb{R}} dw f^*(w)g(w) \quad (5)$	Inner product defining normalization and projection overlaps.	D
$\hat{O}_{\Psi} \equiv \hat{H}_w = -\frac{d^2}{dw^2} + V_{\Psi}(w) \quad (6)$	Internal Hamiltonian / spectral operator.	D
$\hat{O}_{\Psi} u_n\rangle = \mu_n^2 u_n\rangle \quad (7)$	Internal eigenvalue equation.	D
$\langle u_m u_n\rangle = \delta_{mn}, \quad \sum_n u_n\rangle\langle u_n = \mathbf{1} \quad (8)$	Orthonormality and completeness.	D
$S_{\text{MR}} = \int d^4x d\xi \sqrt{-g_{\text{eff}}} \left[\frac{1}{2} G^{AB} D_A \Psi^\dagger D_B \Psi - V_{\text{MR}}(\Psi, \xi) + \mathcal{L}_{\text{src}} \right] \quad (9)$	Master action over spacetime and internal coordinates.	D
$D_A = \partial_A - iqA_A \quad (10)$	Internal/gauge covariant derivative.	D
$J(x) \propto T_{\mu\nu} u^\mu u^\nu, \quad J(x) \propto T^{00}(x) \text{ in the local rest frame} \quad (11)$	Covariant energy-density source rule.	D
$V_{\Psi}(w) = \frac{1}{2}a_2w^2 + \frac{1}{4}a_4w^4 + \dots \quad (12)$	Local internal projection potential expansion.	D/C
$(a_2, a_4, \dots) = \arg \min E_{\text{vac}} \quad (13)$	Vacuum selection principle replacing arbitrary parameter choice.	C

3 Projection Operators and Sector Map

Operator	Physical sector	Tag
$P_g \sim O_G = V_{\Psi}''(w)$	Gravity as internal stiffness projection.	D

Operator	Physical sector	Tag
P_{rot}	Rotation / frame dragging. The rotating extension is treated as an axisymmetric projection-space deformation with frame dragging. / Kerr exterior.	D/C
$P_H(w) = w$	Higgs as displacement projection.	D
P_A and S_θ^1	Gauge field and compact phase projection.	D
P_X	Strong-field continuum / ringdown response.	D/C

4 Emergent Geometry and Weak Gravity

Equation	Meaning	Tag
$Q_{\mu\nu} = \left\langle \partial_\mu \Psi^A \partial_\nu \Psi^B \right\rangle G_{AB} \quad (14)$	Pre-metric projection correlation kernel.	D
$g_{\mu\nu}^{\text{eff}} = \frac{Q_{\mu\nu}}{ \det Q ^{1/4}} \quad (15)$	Determinant-normalized emergent metric.	D
$g_{\mu\nu}^{\text{eff}} = \eta_{\mu\nu} + h_{\mu\nu} + O(q^2) \quad (16)$	Weak-field expansion.	D
$h_{\mu\nu} = q_{\mu\nu} - \frac{1}{4} \eta_{\mu\nu} q, \quad q = \eta^{\alpha\beta} q_{\alpha\beta} \quad (17)$	Trace-subtracted perturbation from determinant normalization.	D
$O_X = -\frac{d^2}{dw^2} + V_X(w) \quad (18)$	Internal projection-manifold operator.	D
$\rho_P(\mu^2) = \sum_n \left \langle u_n \hat{V}_g u_0 \rangle \right ^2 \delta(\mu^2 - \mu_n^2) \quad (19)$	Gravity spectral density.	D
$F(k^2) = \int d\mu^2 \frac{\rho_P(\mu^2)}{k^2 - \mu^2 + i\epsilon} \quad (20)$	Projection self-energy.	D

Equation	Meaning	Tag
	Projection-sector graviton propagator.	D
$D_{\mu\nu\rho\sigma}^{(P)}(k) = \frac{P_{\mu\nu\rho\sigma}^{(2)}}{k^2 - F(k^2) + i\epsilon} \quad (21)$		
	Saturated self-energy; linear at low response and bounded at high response.	D/C
$F_{\text{eff}}(k^2) = \chi k^2 \tanh \left[\frac{F_{\text{raw}}(k^2)}{\chi k^2} \right], \quad 0 < \chi < 1 \quad (22)$		
$F_{\text{eff}}(0) = 0 \quad (23)$	Massless graviton condition.	C
	Projection-corrected spin-2 propagator.	D
$D_{\mu\nu\rho\sigma}^{(P)}(k) = \frac{P_{\mu\nu\rho\sigma}^{(2)}}{k^2 - F_{\text{eff}}(k^2) + i\epsilon} \quad (24)$		
$\frac{F_{\text{eff}}(k^2)}{k^2} \ll 1, \quad k^2 \rightarrow 0 \quad (25)$	GR recovery condition.	C
	Projection-corrected Einstein equation.	D
$G_{\mu\nu}[g_{\text{eff}}] = 8\pi G_P T_{\mu\nu} + \Pi_{\mu\nu}^{(P)} \quad (26)$		
	Low-energy Einstein limit.	D/C
$\Pi_{\mu\nu}^{(P)} \rightarrow 0 \quad \Rightarrow \quad G_{\mu\nu} = 8\pi G_P T_{\mu\nu} \quad (27)$		
	MR Poisson equation.	D
$(k^2 - F(k^2))\Phi(k) = 4\pi G_P \rho(k) \quad (28)$		
	Newtonian limit.	D/C
$\nabla^2 \Phi = 4\pi G_P \rho, \quad \Phi(r) = -\frac{G_P M}{r} \quad (29)$		

5 Stability and Quantum/UV Conditions

Equation	Meaning	Tag
$Z^{-1}(k^2) = 1 - \frac{dF_{\text{eff}}}{dk^2} > 0 \quad (30)$	No-ghost residue condition.	C
$\mu_n^2 > 0 \quad (31)$	No-tachyon internal spectrum condition, apart from controlled symmetry-breaking curvature.	C
	No extra finite- k pole condition.	C
$D^{-1}(k^2) = k^2 - F_{\text{eff}}(k^2) > 0, \quad k^2 > 0 \quad (32)$		
	Stability theorem for saturated self-energy.	D/C
$0 \leq \frac{F_{\text{eff}}(k^2)}{k^2} < \chi < 1 \Rightarrow D^{-1}(k^2) > 0 \quad (33)$		
	Canonical quantization of the MR field.	D
$[\hat{\Psi}(x, \xi), \hat{\Pi}_{\Psi}(y, \xi')] = i\hbar\delta^{(3)}(x-y)\delta(\xi-\xi') \quad (34)$		
	Mode expansion of the quantized projection field.	D
$\hat{\Psi}(x, \xi) = \sum_n \left[a_n(x)u_n(\xi) + a_n^\dagger(x)u_n^*(\xi) \right] \quad (35)$		
	Positivity and UV spectral softening for finite self-energy.	C
$\rho_P(\mu^2) \geq 0, \quad \rho_P(\mu^2) \rightarrow 0 \text{ sufficiently fast as } \mu \rightarrow \infty \quad (36)$		

6 Higgs Projection and Gauge-Higgs Closure

Equation	Meaning	Tag
	Higgs as internal displacement projection.	D
$P_H(w) = w, \quad H(x) = \langle u_0 P_H \Psi(x) \rangle \quad (37)$		

Equation	Meaning	Tag
$Z_H = \sum_n \frac{ \langle u_n w u_0 \rangle ^2}{\mu_n^2} \quad (38)$	Higgs spectral normalization.	D
$V_\Psi(w) = \frac{1}{2}a_2w^2 + \frac{1}{4}a_4w^4 + \dots, \quad a_2 < 0, \quad a_4 > 0 \quad (39)$	Internal projection potential expansion producing symmetry breaking.	D/C
$w_0^2 = -\frac{a_2}{a_4} \quad (40)$	Displaced internal vacuum.	D
$v_{\text{phys}} = \sqrt{Z_H} w_0, \quad v_{\text{phys}}^2 = -\frac{Z_H a_2}{a_4} \quad (41)$	Physical scale from projection displacement.	D/C
$\lambda_H = \frac{a_4}{Z_H^2} \quad (42)$	Quartic closure in the current tester convention.	D
$m_H^2 = -\frac{2a_2}{Z_H} = 2\lambda_H v_{\text{phys}}^2 \quad (43)$	Higgs mass as curvature at the displaced projection vacuum.	D/C
$\ell_P = \lim_{k^2 \rightarrow 0} \frac{F_{\text{eff}}(k^2)}{\chi k^2} \quad (44)$	Low-energy gravitational projection leakage amplitude.	D/C
$\mathcal{P}_{HG} = \ell_P ^2 \quad (45)$	Quadratic projection leakage relation.	D/C

7 Gauge and Fine Structure

Equation	Meaning	Tag
$\Psi = \rho e^{iq\theta}, \quad \theta \sim \theta + 2\pi \quad (46)$	Compact phase sector generating gauge covariance.	D

Equation	Meaning	Tag
	Gauge covariance from local compact phase motion.	D
$D_\mu = \partial_\mu - iqA_\mu, \quad A_\mu \rightarrow A_\mu + q^{-1}\partial_\mu\alpha$ (47)		
	Maxwell tensor and Maxwell equation.	D
$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad \partial_\mu F^{\mu\nu} = J^\nu$ (48)		
	Fine-structure constant from compact gauge normalization.	D/C
$\alpha_{\text{MR}}^{-1} = 4\pi Z_A, \quad Z_A = Z_w Z_\theta C_A$ (49)		
$\Lambda_\theta = 5\pi \left(1 - \frac{4}{5\pi^4}\right)$ (50)	Current compact winding scale.	C
	Compact winding spectrum; factor 2 is sine/cosine quadrature degeneracy.	D/C
$Z_\theta = 2 \sum_{m=1}^{\infty} \exp \left[- \left(\frac{m}{\Lambda_\theta} \right)^3 \right]$ (51)		
	Compact gauge-volume correction from four gauge components plus one compact phase constraint.	D/C
$C_A = \left(1 + \frac{1}{5\pi^4}\right)^{1/4}, \quad N_A = 4+1 = 5$ (52)		
$\alpha_{\text{MR}}^{-1} \simeq 137.1877$ (53)	Current fine-structure result from the corrected compact closure.	C

8 Gauge-Higgs Mass Closure

Equation	Meaning	Tag
$Z_A = Z_2 + Z_1$ (54)	Gauge normalization split used in the reduced closure.	D/C

Equation	Meaning	Tag
$\sin^2 \theta_W^{\text{MR}} = \frac{1}{1 + \pi + 1/(2\pi)} \simeq 0.232518$ (55)	Compact topology split with Higgs-alignment correction.	C
$Z_2 = Z_A \sin^2 \theta_W, \quad Z_1 = Z_A \cos^2 \theta_W$ (56)	Gauge normalization split.	D/C
$g^{-2} = Z_2, \quad g'^{-2} = Z_1, \quad e^{-2} = Z_A$ (57)	Coupling normalizations.	D
$m_W^2 = \frac{v^2}{4Z_2}, \quad m_Z^2 = \frac{v^2}{4} (Z_2^{-1} + Z_1^{-1}), \quad m_\gamma^2 = 0$ (58)	Gauge-Higgs boson masses and exact photon masslessness.	D/C
$C_{EW} = \mathcal{N}_{EW},$ (59)	fixed by the electroweak closure condition.	C
$m_W^{\text{MR}} \simeq 80.116 \text{ GeV}, \quad m_Z^{\text{MR}} \simeq 91.451 \text{ GeV}$ (60)	Current first-order mass closure from compact normalization.	C

9 MR-Native Kerr and Strong Gravity

Equation	Meaning	Tag
$\Psi_{\text{rot}} = R(r)\Theta(\theta)u_0(\xi)e^{-i\omega t + im\phi}$	Rotating MR coherent exterior state.	D/C
$Q_{t\phi} \propto -\omega m R ^2 \Theta ^2$	Native origin of frame dragging from projection phase winding.	D

Equation	Meaning	Tag
$\frac{1}{R} \frac{d}{dr} \left[\Delta_r(r) \frac{dR}{dr} \right] + V_r(r) = -\Lambda,$ $\frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left[\sin \theta \frac{d\Theta}{d\theta} \right] + V_\theta(\theta) = \Lambda$	Separated MR exterior projection equations.	C/O
$\Sigma_{\text{MR}} = r^2 + a^2 \cos^2 \theta$	Angular Kerr structure from axisymmetry, reflection symmetry, smooth nonrotating limit, and no exterior hair.	D/C
$\Delta_{\text{MR}} = r^2 - \frac{2G_P M}{c^2} r + a^2$	Radial Kerr structure from Schwarzschild limit, flat rotating limit, and no extra exterior charge.	D/C
$a = \frac{J}{Mc}$	Kerr spin parameter in physical units.	D
$ds_{\text{MR}}^2 = - \left(1 - \frac{2G_P M r}{c^2 \Sigma_{\text{MR}}} \right) c^2 dt^2$ $- \frac{4G_P M a r \sin^2 \theta}{c^3 \Sigma_{\text{MR}}} dt d\phi$ $+ \frac{\Sigma_{\text{MR}}}{\Delta_{\text{MR}}} dr^2 + \Sigma_{\text{MR}} d\theta^2$ $+ \left(r^2 + a^2 + \frac{2G_P M a^2 r \sin^2 \theta}{c^2 \Sigma_{\text{MR}}} \right) \sin^2 \theta d\phi^2$	MR-derived rotating exterior metric; equals Kerr outside the projection core.	D/C
$r_{\pm} = \frac{G_P M}{c^2} \pm \sqrt{\left(\frac{G_P M}{c^2} \right)^2 - a^2}$	Horizon radii of the exterior Kerr branch.	D/C
$\Psi^3 - \Psi = \frac{\rho}{\rho_*}, \quad R_{\text{max}} \sim \mu_{\text{min}}^2, \quad \rho_* \sim \frac{c^2 R_{\text{max}}}{G_P}$	Nonlinear projection saturation, maximum curvature, and density cutoff.	D/C
$r_c = \left(\frac{G_P M}{c^2 R_{\text{max}}} \right)^{1/3}, \quad \frac{r_c}{r_s} \propto M^{-2/3}$	Finite core radius and scaling relative to Schwarzschild radius.	D/C

10 Gravitational Waves and Ringdown

Equation	Meaning	Tag
$g_{\mu\nu}^{\text{eff}} = \eta_{\mu\nu} + h_{\mu\nu}, \quad h_{\mu\nu} = \delta \left(\frac{\langle \partial_\mu \Psi \partial_\nu \Psi \rangle}{ \det Q ^{1/4}} \right)$ (61)	Gravitational waves are propagating projection-geometry perturbations.	D
$[\square + \Pi_X(\square)] h_{\mu\nu}^{\text{TT}} = 16\pi G_P T_{\mu\nu}^{\text{TT}} \quad (62)$	Native MR gravitational-wave equation.	D/C
$D_{\mu\nu\rho\sigma}^{\text{MR}}(k) = \frac{P_{\mu\nu\rho\sigma}^{(2)}}{k^2 - \Pi_X(k^2) + i\epsilon} \quad (63)$	GW propagator with projection-continuum self-energy.	D
$\Pi_X/k^2 \ll 1 \quad \Rightarrow \quad c_{GW} = c \quad (64)$	Low-energy propagation is GR-like and luminal.	D/C
$A(\omega) = A_{\text{Kerr}}(\omega) + A_X(\omega), \quad A_X(\omega) = \int^{\text{response.}} d\mu \frac{\rho_X(\mu)}{\omega - \mu + i\Gamma_X(\mu)}$ (65)	Kerr response plus internal projection-continuum response.	D/C

11 Action Variation and Field-Equation Checks

Equation	Meaning	Tag
$\mathcal{L}_\Psi = \frac{1}{2} \partial_\alpha \Psi \partial^\alpha \Psi - V(\Psi) \quad (66)$	Scalar projection Lagrangian.	D
$\frac{\partial \mathcal{L}}{\partial \Psi} - \partial_\alpha \left(\frac{\partial \mathcal{L}}{\partial (\partial_\alpha \Psi)} \right) = 0 \quad (67)$	Euler–Lagrange equation.	D
$\square \Psi + V'(\Psi) = 0 \quad (68)$	Field equation from the scalar projection action.	D

Equation	Meaning	Tag
	Gauge-field action with projection normalization.	D
$\mathcal{L}_A = -\frac{1}{4}Z_A F_{\mu\nu} F^{\mu\nu} + J_\mu A^\mu \quad (69)$		
	Gauge equation from variation.	D
$Z_A \partial_\mu F^{\mu\nu} = J^\nu \quad (70)$		
	Quadratic action kernel and propagator inverse relation.	D
$K(k) = k^2 - F(k^2), \quad D(k) = K(k)^{-1} \quad (71)$		

12 Hubble Expansion Sector

The projection-sector cosmological closure is written in terms of the observable expansion function $H(z)$:

$$H(z)^2 = H_0^2 \left[\Omega_m(1+z)^3 + \Omega_r(1+z)^4 + \Omega_\theta(z) \right]. \quad (72)$$

Here $\Omega_\theta(z)$ is the protected homogeneous projection-sector contribution. It represents the slow, near-zero projection mode responsible for late-time expansion while remaining decoupled from the fast local gravity, Kerr, Higgs, and spectral-response sectors.

13 Observable Benchmarks and Test Status

Quantity	Current MR result	Status
Fine structure	$\alpha_{\text{MR}}^{-1} \simeq 137.1877$	Strong first-order; $\sim 0.11\%$
Weak angle	$\sin^2 \theta_W^{\text{MR}} \simeq 0.232518$	Strong first-order
Gauge-Higgs masses	$m_W^{\text{MR}} \simeq 80.116 \text{ GeV}, m_Z^{\text{MR}} \simeq 91.451 \text{ GeV}$	Sub-percent first-order
Weak gravity	GR/Newton recovered for $F/k^2 \ll 1$	Strong
Kerr	MR-native exterior closure with remaining full Ricci symbolic run pending	Strong framework
Finite core	$r_c/r_s \sim 10^{-21}$ for $65M_\odot$ benchmark	Passed toy regularity test
GW propagation	Low-energy propagation recovers GR form	Framework closed

14 Validation Tester Suite

Tester	Purpose	Output File
Symbolic + numerical tester v1	Algebraic closures, gauge–Higgs identities, compact numerical scoreboard	master_relativity_symbolic_numeric_tester_v1_results.csv
Tensor consistency tester v2.1	Kerr determinant, inverse, limits, Schwarzschild Ricci-flatness	master_relativity_tensor_consistency_tester_v2_1_results.csv
Action variation tester v3.1	Euler–Lagrange checks, Higgs vacuum, gauge variation, propagator kernel	master_relativity_action_variation_tester_v3_1_results.csv
Curvature geometry tester v4	Schwarzschild Ricci scalar/tensor, Einstein tensor, Kerr algebraic geometry	master_relativity_curvature_geometry_tester_v4_results.csv
Full Kerr Ricci tester v4.3	Full symbolic rotating Kerr Ricci-flatness check	master_relativity_full_kerr_ricci_v4_3_results.csv
Semi-symbolic Kerr Ricci tester v4.4	Faster Kerr Ricci check with $G = M = 1$, $a = 1/2$	master_relativity_semi_symbolic_kerr_ricci_v4_4_results.csv
Finite-core regularity tester v5.1	Saturation, bounded curvature, core scaling, weak-curvature recovery	master_relativity_finite_core_tester_v5_1_results.csv
Weak-field observational tester v6	Mercury precession, light bending, Shapiro delay, weak-field compactness	master_relativity_weak_field_tester_v6_results.csv
Compact/electroweak lock tester v7	Higgs normalization, fine-structure, weak angle, W/Z closure	master_relativity_compact_electroweak_lock_tester_v7_results.csv
Lorentz speed closure tester v8.1	Universal propagation speed from equal temporal/spatial stiffness	master_relativity_lorentz_speed_closure_tester_v8_1_results.csv
GW propagation/stability tester v9	Luminal GW recovery, no ghosts, no extra poles, stable continuum denominator	master_relativity_gw_propagation_stability_tester_v9_results.csv

15 Parameter Audit

Parameter	Meaning	Tag	Comment
Ψ	Master projection field	D	fundamental
$\hat{O}_\Psi$	Spectral operator	D	internal Hamiltonian

Parameter	Meaning	Tag	Comment
V_Ψ	Projection potential	C/O	selected by E_{vac}
u_n, μ_n	Spectrum	D	eigen-system
$Q_{\mu\nu}$	Metric seed	D	bilinear correlation
$g_{\mu\nu}^{\text{eff}}$	Observable metric	D	determinant normalized
G_P	Newton coupling	C	fixed by low-energy gravity
$F(0) = 0$	Massless graviton	C	required for GR recovery
R_{max}	Maximum curvature	C/O	linked to spectral cutoff
r_c	Finite core radius	D/C	follows from saturation
$\Sigma_{\text{MR}}, \Delta_{\text{MR}}$	Kerr exterior functions	D/C	no-hair/separability assumptions
Z_H	Higgs normalization	D/C	spectral overlap
a_2, a_4	Higgs potential curvature/stiffness	D/C	from internal potential
Z_w, Z_θ, C_A	Fine-structure factors	D/C	compact gauge geometry
$p = 3$	Winding suppression exponent	C/O	compact-volume closure, not uniqueness theorem
Λ_θ	Compact winding scale	C/O	current corrected compact closure
C_{EW}	Gauge-Higgs normalization	C	compact correction
Π_X	GW self-energy	D/C	continuum correction

16 Current Completion Status

- Strongly closed at framework level: master operator, projection metric, weak gravity, Higgs potential, gauge/Maxwell, fine structure, gauge-Higgs first-order masses, finite-core scaling, gravitational-wave propagation.
- Strong but with theorem-level caveats: MR-native Kerr uniqueness, compact winding exponent unique-

ness, gauge-Higgs alignment correction.

- Current validation suite: symbolic algebra, tensor identities, action variation, Schwarzschild curvature, finite core regularity, weak-field observations.
- Open frontier: full rotating Kerr Ricci symbolic completion, exact compact-winding theorem, full UV proof beyond EFT-level closure.